Probability Theory

Labix

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Abstract

Notes for the basics of Probability Theory.

Contents

1 Foundations of Probability Theory

1.1 Definition of Probability

Definition 1.1.1 (Probability Space)
A probability space is a measure space $(\Omega, \mathcal{F}, P : \mathcal{F} \to [0,1])$.

Explicitly, a probability space is a triple $(\Omega, \mathcal{F}, P)$ consisting of the following data:

- $\Omega \neq \emptyset$ is a set called the sample space.
- $\mathcal{F} \subseteq \mathcal{P}(\Omega)$ is a σ-algebra called events.
- $P : \mathcal{F} \to [0,1]$ is a set function.

such that the following are true:

- $P(\Omega) = 1$.
- If $\{A_n \mid n \in \mathbb{N}\} \subseteq \mathcal{F}$ are pairwise disjoint, then

$$P\left(\bigcup_{k=1}^{\infty} A_k\right) = \sum_{k=1}^{\infty} P(A_k)$$

Definition 1.1.2 (Events and Probability)
Let $(\Omega, \mathcal{F}, P)$ be a probability space.

- An event of the space is an element of $\mathcal{F}$.
- Let $A \in \mathcal{F}$ be an event. The probability of A is defined to be the value $P(A)$.

Example 1.1.3 (Uniform Probability)
Let Ω be the set of all possible outcomes of flipping a coin ten times. Let $\mathcal{F} = \mathcal{P}(\Omega)$ be the set of subsets of Ω. Let $P : \mathcal{F} \to [0,1]$ be the function defined by $P(E) = \frac{|E|}{|\Omega|}$ for all $E \in \mathcal{F}$. Then $(\Omega, \mathcal{F}, P)$ is a probability space.

Proposition 1.1.4
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $A, B \in \mathcal{F}$ be events. Then the following are true.

- $P(\Omega \setminus A) = 1 - P(A)$
- $A \subset B \implies P(A) \leq P(B)$

Proof
Let $A \subset B \subset \Omega$ be events in Ω.

- A and $\Omega \setminus A$ are disjoint and $P(\Omega) = P(A) + P(\Omega \setminus A)$ and $P(\Omega \setminus A) = 1 - P(A)$
- We have that A and $B \setminus A$ are disjoint. Thus $P(B) = P(A) + P(B \setminus A)$. Since $P(B \setminus A) \geq 0$, we have $P(A) \leq P(B)$.

$\blacksquare$

Example 1.1.5 (The Birthday Problem)
Suppose that there are k people in a room, all of whom are equally likely to have a birthday in one of the 365 days in a year. Suppose that the birthdays of the people in the room are independent of each other. What is the probability that two or more people in the room have the same birthday?

Answer
Let $\Omega = \{\{n_1, \dots, n_k\} \mid 1 \leq n_1, \dots, n_k \leq 365\}$. These indicate all the possible birthday configurations. Let $\mathcal{F} = \mathcal{P}(\Omega)$. This means that every birthday configuration we constructed is possible.

By the assumptions of the question, we set $P(E) = \frac{|E|}{|\Omega|}$ for an event $E \in \mathcal{F}$. Then we have

$$P(\text{at least one birthday match}) = 1 - P(\text{no birthday matches}) = 1 - \frac{365!}{(365)^k(365-k)!}$$

⊛

Theorem 1.1.6 (Principle of Inclusion Exclusion)
Let $A, B \subset \Omega$ be a sample space and P the probability measure.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Proof
Note that

$$\begin{aligned} A \cup (B \setminus A) &= A \cup (B \cap A^c) \\ &= (A \cup B) \cap (A \cup A^c) \\ &= A \cup B \end{aligned}$$

Note also that $A \cap (B \setminus A) = \emptyset$. Thus $P(A \cup B) = P(A) + P(B \setminus A) = P(A) + P(B) - P(A \cap B)$ ■

Theorem 1.1.7 (Extended Principle of Inclusion Exclusion) Let $A_k \subset \Omega$ be a sample space and P the probability measure for all $k \leq n \in \mathbb{N}$. Then

$$P\left(\bigcup_{k=1}^{n} A_k\right) = \sum_{k=1}^{n} (-1)^{k+1} \sum_{1 \leq i_1 < \cdots < i_k \leq n} P(A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k})$$

Example 1.1.8
A casino offers a card game with the following rules. You draw a card from a deck of cards. The host then draws a card from the deck without replacement. If your card is strictly larger than the card your host holds, you win. What is your probability of winning? (c.f. Green Book of Quant)

Answer
We could have used the law of total probability (refer to later sections), but we could just use logic. Let E_1 be the event that your card has strictly larger value than the host. Let E_2 be the converse. Let E_3 be the event that both card has the same value. Since E_1, E_2, E_3 are pairwise disjoint events that covers all possible outcomes, we have $P(E_1) + P(E_2) + P(E_3) = 1$ by the principle of inclusion exclusion. Since E_1 and E_2 are symmetric, we have $P(E_1) = P(E_2)$. Finally, notice that $P(E_3) = \frac{3}{51} = \frac{1}{17}$. Hence we conclude that $P(E_1) = \frac{8}{17}$. ⊛

Example 1.1.9
A line of 100 airline passengers are waiting to board a plane, and they each hold a ticket to one of the 100 seats on that flight; for convenience, let's say that the nth passenger in line has a ticket for seat number n. Being drunk, the first person in line picks a random seat, equally likely for each seat. All of the other passengers are sober and will go to their proper seats unless it is already occupied; in that case, they will randomly choose a free seat. You're person number 100. What is the probability that you end up in your seat (i.e., seat 100)? (c.f. Green Book of Quant)

Answer
Let E_1 be the event that seat 100 is taken before seat 1. Let E_2 be the event that seat 1 is taken before seat 100. I claim that E_2 is equal to the event that you take your own seat. Indeed, since you are the last passenger to board, if you take your own seat, then seat 1 must have already been taken. Now suppose that seat 1 is taken before seat 100 and suppose that the first passenger has taken seat n. Then passengers 2 to n-1 will have their own seat, and passenger n either randomly takes seat 1 or any other seat. If he takes seat 1, then passengers n+1 to 99 will seat at their own

seats and hence you will seat at your own seat. If he does not take seat 1, and takes seat m instead for $100 > m > n$, then the process repeats until seat 1 is taken before seat 100 or $m = 99$, in which case by assumption passenger 99 must take seat 1 and you are left with seat 100.

Now E_1 and E_2 are disjoint events that covers all possible outcomes and are symmetric. By the principle of inclusion exclusion, we conclude that $P(E_2) = \frac{1}{2}$. ⊛

Example 1.1.10
Given N points on a circle, what is the probability that they all lie in a semicircle? (c.f. Green Book of Quant)

1.2 Conditional Probability

Definition 1.2.1 (Conditional Probability)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $A, B \in \mathcal{F}$ be events. Define the conditional probability of A given B to be

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Example 1.2.2 (Two Children Problem)

Example 1.2.3
A family has two children. Given that one of them is a boy, what is the probability that both children are boys.

Answer
We use the sample space $\Omega = \{(b,b),(b,g),(g,b),(g,g)\}$ and $\mathcal{F} = \mathcal{P}(\Omega)$ for the space of events. We assume uniform probability in the four cases. Let A be the event that both children are boys. Let B be the event that one of the child is a boy. Then we have

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{P(\{(b,b)\})}{P(\{(b,b),(b,g),(g,b)\})} = \frac{1}{3}$$

⊛

Proposition 1.2.4 (Multiplication Rule)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $A_1, \dots, A_n \in \mathcal{F}$ be events. Then we have

$$P\left(\bigcap_{i=1}^n A_i\right) = P(A_1) \cdot \prod_{i=2}^n P\left(A_i \,\middle|\, \bigcap_{j=1}^{i-1} A_j\right)$$

Proof
Follow immediately by induction on the definition of conditional probability. ■

Example 1.2.5
Suppose there are five buckets whose labels (1,2,3,4,5) are covered up. Suppose that there are five balls with the same labels. A ball is placed randomly into each bucket. What is the probability that none of the labels of the balls match the labels on the buckets? (c.f. Green Book of Quant)

Answer
Let E_i be the event that the ith ball is placed correctly into the ith bucket. By the principle of

inclusion exclusion, we have

$$P\left(\bigcup_{i=1}^{5} E_i\right) = \sum_{k=1}^{5}(-1)^{k+1}\sum_{1\leq i_1<\cdots<i_k\leq 5} P(E_{i_1}\cap\cdots\cap E_{i_k})$$

When $k=1$, since we assume that the random placement of the ball is uniform, we have $P(E_i)=\frac{1}{5}$. When $2\leq k\leq 5$, by the multiplication rule, we deduce that

$$P(E_{i_1}\cap\cdots\cap E_{i_k}) = P(E_{i_1})\prod_{j=2}^{k} P\left(E_{i_j}\,\middle|\,\bigcap_{t=1}^{j-1} E_{i_t}\right) = \frac{1}{5}\prod_{j=2}^{k}\frac{1}{6-j} = \frac{(5-k)!}{5!}$$

Combining with the principle of inclusion exclusion, we have

$$P\left(\bigcup_{i=1}^{5} E_i\right) = \sum_{k=1}^{5}(-1)^{k+1}\binom{5}{k}\frac{(5-k)!}{5!} = \sum_{k=1}^{5}(-1)^{k+1}\frac{1}{k!} = \frac{19}{30}$$

So the probability that no labels of the balls match the labels on the bucket is $\frac{11}{30}$. ⊛

Theorem 1.2.6 (Bayes' Theorem)
Let $(\Omega,\mathcal{F},P)$ be a probability space. Let $A,B\in\mathcal{F}$ be spaces such that $P(A),P(B)>0$. Then we have

$$P(A\mid B) = \frac{P(B\mid A)P(A)}{P(B)}$$

Proof
We have that $P(A\cap B)=P(A\mid B)P(B)$ and $P(A\cap B)=P(B\mid A)P(A)$. ■

Theorem 1.2.7 (Law of Total Probability)
Let $(\Omega,\mathcal{F},P)$ be a probability space. Let $A_1,\dots,A_n\in\mathcal{F}$ be events that partition Ω such that $P(A_i)>0$ for $1\leq i\leq n$. Let $B\in\mathcal{F}$ be an event. Then we have

$$P(B) = \sum_{k=1}^{n} P(A_k)P(B\mid A_k)$$

Proof
Note that since $A_1,\dots,A_n$ is a partition of Ω, $B\cap A_1,\dots,B\cap A_n$ is a partition of B. Then we have

$$\begin{aligned}\sum_{k=1}^{n} P(A_k)P(B\mid A_k) &= \sum_{k=1}^{n} P(B\cap A_k)\\ &= P\left(\bigcup_{k=1}^{n} B\cap A_k\right)\\ &= P(B\cap\Omega)\\ &= P(B)\end{aligned}$$

■

Example 1.2.8
Suppose that a coin tossing game is played among person A and B. If A has $n+1$ fair coins and B has n fair coins, what is the probability that A has more heads than B if they both flip all their coins? (c.f. Green Book of Quant)

Answer
Let $\Omega=\{(i,j)\in\mathbb{N}^2\mid 0\leq i\leq n+1, 0\leq j\leq n\}$ be the sample space. Let $\mathcal{F}=\mathcal{P}(\Omega)$ be the set of

events. Let $P : \mathcal{F} \to [0,1]$ be the uniform probability on Ω. Then we have

$$P(\text{A has more heads than B}) = \sum_{i=0}^{n} P(\text{A has more heads than B} \mid B \text{ has } i \text{ heads})P(\text{B has i heads})$$

by the law of total probability. We deduce that

$$P(\text{A has more heads than B}) = \sum_{i=0}^{n} \frac{n-i+1}{n+2}\frac{1}{n+1} = \frac{1}{(n+2)(n+1)}\frac{(n+2)(n+1)}{2} = \frac{1}{2}$$

⊛

Example 1.2.9 (Gambler's Ruin)
A gambler starts with an initial fortune of £k for $1 \leq k \leq n$. On each successive gamble, the gambler either wins £1 or loses £1 independent of past games, with a probability of p and $1-p$ respectively for $p \in (0,1)$. What is the probability that the gambler reaches £n dollars before reaching £0?

Answer
The gambler is said to have lost the game if the gambler reaches £0 after any round. Let a_i denote the probability that the gambler reaches £n dollars before reaching £0, when the gambler has £i. By the law of total probability, we obtain a recurrence relation

$$a_i = pa_{i+1} + (1-p)a_{i-1}$$

This makes for a difference equation with boundary conditions $a_0 = 1$ and $a_n = 0$. Solving it gives

$$a_i = \begin{cases} \frac{i}{n} & \text{if } p = \frac{1}{2} \\ \frac{1-((1-p)/p)^i}{1-((1-p)/p)^n} & \text{otherwise} \end{cases}$$

Notice that the $p = 1/2$ case is consistent with the $p \neq 1/2$ cases since $\lim_{p\to 1/2} \frac{1-((1-p)/p)^i}{1-((1-p)/p)^n} = \frac{i}{n}$.
⊛

Example 1.2.10
You are given 1000 coins. Among them, 1 coin has heads on both sides. The other 999 coins are fair coins. You randomly choose a coin and toss it 10 times. Each time, the coin turns up heads. What is the probability that the coin you choose is the unfair one? (c.f. Green Book of Quant)

Answer
Let A be the event that the chosen coin was unfair. Let B be the event that the coin turns up heads ten times. Using Bayes' theorem and the law of total probability, we get

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)} = \frac{P(A)}{P(A)P(B \mid A) + P(A^c)P(B \mid A^c)} = \frac{1/1000}{1/1000 + 999/1000 \times (1/2)^{10}}$$

which can be evaluated into $\frac{1024}{2023}$. ⊛

Proposition 1.2.11
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $B \in \mathcal{F}$ be an event such that $P(B) > 0$. Then $(\Omega, \mathcal{F}, P(- \mid B))$ is a probability space.

1.3 Independent Events

Definition 1.3.1 (Independent Events)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $A, B \in \mathcal{F}$ be events. We say that A and B are independent if

$$P(A \cap B) = P(A)P(B)$$

Proposition 1.3.2
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $A, B \in \mathcal{F}$ be events. If A and B are independent, then the following are true.

- $\Omega \setminus A$ and B are independent.
- A and $\Omega \setminus B$ are independent.
- $\Omega \setminus A$ and $\Omega \setminus B$ are independent.

2 Probability Distributions

2.1 Random Variables and its Distribution

Definition 2.1.1 (Random Variable)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $(E, \mathcal{E})$ be a measurable space. An $(E, \mathcal{E})$ valued random variable is an $\mathcal{F}$-measurable function $X : \Omega \to E$.

Definition 2.1.2 (Discrete and Continuous Random Variables)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}$ be a random variable.
- We say that X is discrete if $\text{im}(X)$ is a countable subset of $\mathbb{R}$.
- We say that X is continuous otherwise.

Recall that X is an $\mathcal{F}$-measurable function if $X^{-1}(B) \in \mathcal{F}$ for $B \in \mathcal{E}$.

Example 2.1.3
Let Ω be the the sample space of tossing three fair coins. Let $X : \Omega \to \mathbb{R}$ be the number of heads observed. Let $Y : \Omega \to \mathbb{R}$ be the number of tails observed. Then X and Y are identically distributed, but $X \neq Y$.

Definition 2.1.4 (Probability Density Functions and Probability Mass Function)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}$ be a random variable.
- Suppose that X is a discrete random variable. Define the probability mass function of X to be the Radon–Nikodym derivative
$$f_X = \frac{dX_*P}{d\#}$$
where $\#$ is the counting measure on $\mathbb{R}$.
- Suppose that X is a continuous random variable. Define the probability density function of X to be the Radon–Nikodym derivative
$$f_X = \frac{dX_*P}{d\mu}$$
where μ is the Lebesgue measure.

Recall that this means that f_X satisfies the property that
$$P_X(A) = \int_A f_X \, d\mu$$
for any measurable set $A \subseteq \mathbb{R}$. In particular, if $A = \{a\} \subseteq \mathcal{F}$, then we have
$$f_X(a) = P_X(a)$$
The probability distribution function has its input as every measurable subset of $\mathbb{R}$, while the probability density function takes input as individual points of $\mathbb{R}$. They are really the same thing because having its probability be determined on singletons is sufficient to determine the probability of every measurable subset.

Proposition 2.1.5
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $f : X \to \mathbb{R}$ be a function. Then the following are true.
- f is the probability mass function of a discrete random variable if and only if $f \geq 0$ and $\sum_{x \in X} f_X = 1$.
- f is the probability density function of a continuous random variable if and only if $f \geq 0$ and $\int_{-\infty}^{\infty} f \, dx = 1$.

Proposition 2.1.6
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}$ be a random variable. Let $g : \mathbb{R} \to \mathbb{R}$ be a function. Then the following are true.

- Suppose that X is a discrete random variable. Then the probability density function of $Y = g \circ X$ is given by

$$f_Y(y) = \begin{cases} \sum_{x \in g^{-1}(y)} f_X(x) & \text{if } y \in \text{im}(g) \\ 0 & \text{otherwise} \end{cases}$$

- Suppose that X is a continuous random variable. Suppose that g is strictly monotone and differentiable. Then the probability density function of $Y = g \circ X$ is given by

$$f_Y(y) = \begin{cases} f_X(g^{-1}(y)) \left| \frac{d}{dy}(g^{-1}(y)) \right| & \text{if } y \in \text{im}(g) \\ 0 & \text{otherwise} \end{cases}$$

2.2 Cumulative Density Functions

Definition 2.2.1 (Cummulative Distribution Function)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}$ be a random variable. Define the cummulative distribution function $F_X : \mathbb{R} \to \mathbb{R}$ of X to be

$$F_X(x) = P_X(X \leq x)$$

Proposition 2.2.2
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}$ be a random variable. Then the following are true.

- $f_X = \frac{dF_X}{dx}$.
- $F_X(x) = \int_{-\infty}^{x} f_X(t) \, dt$

Proposition 2.2.3
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}$ be a random variable. Then the following are true regarding the cumulative distribution function F_X.

- F_X is monotonically increasing: $x \leq y \implies F_X(x) \leq F_X(y)$
- F_x is right continuous: If (x_n) is a sequence such that $x_1 \geq \cdots \geq x_n \geq x_{n+1} \geq \cdots \geq x$ and $(x_n) \to x$, then $F_X(x_n) \to F_X(x)$
- $F_X(-\infty) = 0$ and $F_X(\infty) = 1$

Proposition 2.2.4
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}$ be a random variable. Then we have

$$P(\{\omega \in \Omega \mid a < X(\omega) \leq b\}) = F_X(b) - F_X(a)$$

2.3 Independent Variables and Identical Distributions

Definition 2.3.1 (Independent Random Variables)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $(E, \mathcal{E})$ be a measurable space. Let $X, Y : \Omega \to E$ be random variables. We say that X and Y are independent if for any $A, B \in \mathcal{E}$, we have that $X^{-1}(A)$ and $Y^{-1}(B)$ are independent events in $\mathcal{F}$.

Definition 2.3.2 (Probability Distribution)
Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Let $(E, \mathcal{E})$ be a measurable space. Let $X : \Omega \to E$ be a measurable function. Define the probability distribution of X to be the pushforward measure

$P \circ X^{-1} = P_X : \mathcal{E} \to [0, 1]$ defined by

$$P_X(A) = P(X^{-1}(A))$$

for $A \in \mathcal{E}$.

Definition 2.3.3 (Identical Distributions)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X, Y : \Omega \to \mathbb{R}$ be random variables. We say that X and Y are identically distributions if $P_X = P_Y$.

Lemma 2.3.4
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X, Y : \Omega \to \mathbb{R}$ be random variables. Then the following are equivalent.
- X and Y are identically distributed.
- $f_X = f_Y$.
- $F_X = F_Y$.

2.4 Examples of Discrete Random Variables

Definition 2.4.1 (Discrete Uniform Distribution)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}$ be a random variable. We say that X has a discrete uniform distribution if the probability density function of X is given by

$$f_X(x) = \begin{cases} \frac{1}{b-a+1} & \text{if } x \in \mathbb{Z} \text{ and } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

for some $a \leq b$. In this case, we write $X \sim \text{Uniform}(a, b)$

Definition 2.4.2 (Bernoulli Distribution)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}$ be a random variable. We say that X has a Bernoulli distribution if the probability density function of X is given by

$$f_X(x) = \begin{cases} p & \text{if } x = 1 \\ 1-p & \text{if } x = 0 \\ 0 & \text{otherwise} \end{cases}$$

for some $p \in [0, 1]$. In this case we write $X \sim \text{Bern}(p)$.

Definition 2.4.3 (Binomial Distribution)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}$ be a random variable. We say that X has a binomial distribution if the probability density function of X is given by

$$f_X(x) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x} & \text{if } x = 0, \dots, n \\ 0 & \text{otherwise} \end{cases}$$

for some $p \in [0, 1]$ and $n \in \mathbb{N}$. In this case we write $X \sim \text{Binom}(n, p)$.

Definition 2.4.4 (Poisson Distribution)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}$ be a random variable. We say that X has a Poisson distribution if the probability density function of X is given by

$$f_X(x) = \begin{cases} \frac{\lambda^x}{x!} e^{-x} & \text{if } x \in \mathbb{N} \\ 0 & \text{otherwise} \end{cases}$$

for some $\lambda > 0$. In this case we write $X \sim \text{Poisson}(\lambda)$.

Definition 2.4.5 (Geometric Distribution)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}$ be a random variable. We say that X has a geometric distribution if the probability density function of X is given by

$$f_X(x) = \begin{cases} p(1-p)^x & \text{if } x \in \mathbb{N} \\ 0 & \text{otherwise} \end{cases}$$

for some $p \in [0, 1]$. In this case we write $X \sim \text{Geom}(p)$.

2.5 Examples of Continuous Random Variables

Let $I \subseteq \mathbb{R}$ be an interval. Recall that $\mathcal{B}(I)$ refers to the borel measurable subsets of I. Denote λ the Lebesgue measure on $\mathbb{R}^n$.

Definition 2.5.1 (Continuous Uniform Distribution)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to [a, b]$ be a random variable for some $a, b \in \mathbb{R}$. We say that X has a continuous uniform distribution if the probability density function of X is given by

$$f_X(x) = \frac{1}{b-a}$$

for all $x \in [a, b]$.

Definition 2.5.2 (Normal Distribution)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}$ be a random variable. We say that X has a normal distribution if the probability density function of X is given by

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

for some $\mu, \sigma \in \mathbb{R}$ such that $\sigma > 0$. In this case we write $X \sim \text{Normal}(\mu, \sigma)$.

Definition 2.5.3 (Exponential Distribution)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}$ be a random variable. We say that X has an exponential distribution if the probability density function of X is given by

$$f_X(x) = \begin{cases} \frac{1}{\lambda} e^{-x/\lambda} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

for some $\lambda > 0$. In this case we write $X \sim \text{Exp}(\lambda)$.

Definition 2.5.4 (Gamma Distribution)

2.6 Algebra of Random Variables

Proposition 2.6.1
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X, Y : \Omega \to \mathbb{R}$ be random variables. Then we have

$$f_{X+Y}(z) = \int_{-\infty}^{\infty} f_X(w) f_Y(z-w) \, dw$$

3 Expectation and Variance

3.1 Expectations

Definition 3.1.1 (Expectations)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}$ be a random variable. Define the expectation of X to be

$$E[X] = \int_{\Omega} X dP$$

Lemma 3.1.2
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}$ be a random variable. Then we have

$$E[X] = \int_{-\infty}^{\infty} x f_X(x)\, dx$$

Lemma 3.1.3
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $A \in \mathcal{F}$ be an event. Then we have

$$E[1_A] = P(A)$$

Example 3.1.4
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}$ be a random variable such that $X \sim \text{Binom}(n, p)$ for some $n \in \mathbb{N}$ and $p \in [0, 1]$. Then we have

$$E[X] = np$$

Proof
We have

$$\begin{aligned}
E[X] &= \sum_{k=0}^{n} k \binom{n}{k} p^j (1-p)^{n-k} \\
&= \sum_{k=1}^{n} \frac{n! \cdot k}{k!(n-k)!} p^j (1-p)^{n-k} \\
&= \sum_{k=1}^{n} n \binom{n-1}{k-1} p^j (1-p)^{n-k} \\
&= \sum_{i=0}^{n-1} n \binom{n-1}{i} p^{i+1} (1-p)^{n-i-1} && (i = k-1) \\
&= np \sum_{i=0}^{n-1} \binom{n-1}{i} p^i (1-p)^{n-i-1} \\
&= np
\end{aligned}$$

■

Example 3.1.5
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}$ be a random variable such that $X \sim \text{Exp}(\lambda)$ for some $\lambda > 0$. Then we have

$$E[X] = \lambda$$

Proof
We have

$$\begin{aligned} E[X] &= \int_0^\infty \frac{1}{\lambda} x e^{-x/\lambda}\, dx \\ &= [-xe^{-x/\lambda}]_0^\infty + \int_0^\infty e^{-x/\lambda}\, dx \\ &= \int_0^\infty e^{-x/\lambda}\, dx \\ &= \lambda \end{aligned}$$

■

Proposition 3.1.6 Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X, Y : \Omega \to \mathbb{R}$ be random variables. Then the following are true.

- If X, Y are random variables and $a, b \in \mathbb{R}$, then
$$E[aX + bY] = aE[X] + bE[Y]$$
- If $P(X \geq Y) = 1$, then
$$E[X] \geq E[Y]$$

Proposition 3.1.7 Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X, Y : \Omega \to \mathbb{R}$ be random variables. Then X, Y are independent if and only if

$$E[g(X)h(Y)] = E[g(X)]E[h(Y)]$$

for any two functions $g, h : \mathbb{R} \to \mathbb{R}$.

3.2 Variance and Covariance

Definition 3.2.1 (Variance)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}$ be a random variable. Define the variance of X to be
$$\text{Var}(X) = E[(X - E[X])^2]$$

Example 3.2.2
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}$ be a random variable such that $X \sim \text{Exp}(\lambda)$. Then we have
$$\text{Var}(X) = \lambda^2$$

Lemma 3.2.3
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}$ be a random variable. Then the following are true.

- $\text{Var}(X) \geq 0$.
- $\text{Var}(X) = 0$ if and only if $P_X(E[X]) = 1$.
- $\text{Var}(X) = E[X^2] - E[X]^2$
- $\text{Var}(aX + b) = a^2\text{Var}(X)$ for any $a, b \in \mathbb{R}$.

Definition 3.2.4 (Covariance)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X, Y : \Omega \to \mathbb{R}$ be random variables. Define the covariance of X and Y to be
$$\text{Cov}(X, Y) = E[(X - E(X))(Y - E(Y))]$$

Lemma 3.2.5
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}$ be a random variable. Then we have
$$\text{Cov}(X, X) = \text{Var}(X)$$

Proposition 3.2.6
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X, Y : \Omega \to \mathbb{R}$ be random variables. Then the following are true.

- $\text{Cov}(X, Y) = \text{Cov}(Y, X)$.
- $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$.
- If X, Y are independent, $\text{Cov}(X, Y) = 0$.

Lemma 3.2.7
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X, Y, Z : \Omega \to \mathbb{R}$ be random variables. Then we have
$$\text{Cov}(aX + bY, Z) = a\text{Cov}(X, Z) + b\text{Cov}(Y, Z)$$
for any $a, b \in \mathbb{R}$.

Proposition 3.2.8
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \to \mathbb{R}$ be random variables. Then we have
$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}\left(X_i\right) + 2 \sum_{1 \le i < j \le n} \text{Cov}\left(X_i, X_j\right)$$

Proposition 3.2.9
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X, Y : \Omega \to \mathbb{R}$ be random variables. Then we have
$$|\text{Cov}(X, Y)| \le \sqrt{\text{Var}(X)\text{Var}(Y)}$$

3.3 Moments and Moment Generating Functions

Definition 3.3.1 (Moments)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}$ be a random variable. Define the nth moment of X to be
$$\mu_n = E[X^n]$$

Definition 3.3.2 (Moment Generating Function)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}$ be a random variable. Define the moment generating function of X to be the function $M_X : \mathbb{R} \to \mathbb{R}$ given by
$$M_X(t) = E[e^{tX}]$$
whenever the expectation is defined.

Proposition 3.3.3 Assume that M_X exists in a neighbourhood of 0, that is, there exists $\epsilon > 0$ such that for all $t \in (-\epsilon, \epsilon)$ we have $M_X(t) < \infty$. Then for all $k \in \mathbb{N}$ the kth moment of X exists, and
$$E[X^k] = \frac{d^k}{dt^k} M_X(t)\bigg|_{t=0}$$

Proof We have that $E[X^k] = \int_{-\infty}^{\infty} x^k f_X(x)\,dx$ for any continuous cummulative probability. On the other hand,

$$\begin{aligned}\frac{d^k}{dt^k} M_X(t)\bigg|_{t=0} &= \frac{d^k}{dt^k} \int_{-\infty}^{\infty} e^{tx} f_X(x)\,dx\bigg|_{t=0} \\ &= \int_{-\infty}^{\infty} \frac{\partial^k}{\partial t^k} e^{tx} f_X(x)\,dx\bigg|_{t=0} \\ &= \int_{-\infty}^{\infty} x^k e^{tx} f_X(x)\,dx\bigg|_{t=0} \\ &= \int_{-\infty}^{\infty} x^k f_X(x)\,dx\end{aligned}$$

■

Proposition 3.3.4
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}$ be a random variable. Then the following are true regarding the moment generating function.

- For any $a, b \in \mathbb{R}$, we have $M_{aX+b}(t) = e^{tb} M_X(at)$
- If X, Y are independent, then $M_{X+Y} = M_X \cdot M_Y$

Proposition 3.3.5 Let X, Y be two random variables. Assume that the moment generating functions of X, Y exists and are finite on an interval of the form $(-\epsilon, \epsilon)$. Assume further that $M_X(t) = M_Y(t)$ for all $t \in (-\epsilon, \epsilon)$. Then X, Y have the same distribution.

4 Multivariate Probability Theory

4.1 Multiple Random Variables

Definition 4.1.1 (Joint Probability Density and Mass Functions)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X, Y : \Omega \to \mathbb{R}$ be random variables.

- Suppose that X and Y are discrete random variables. Define the joint probability mass function of X and Y to be the Radon-Nikodym derivative
$$f_{X,Y} = \frac{d(X,Y)_*P}{d\#}$$
where $\#$ is the counting measure on $\mathbb{R}^2$.
- Suppose that X and Y are continuous random variables. Define the joint probability density function of X and Y to be the Radon-Nikodym derivative
$$f_{X,Y} = \frac{d(X,Y)_*P}{d\mu^2}$$
where μ^2 is the Lebesgue measure on $\mathbb{R}^2$.

As with the case of pmfs with one random variable, we have
$$f_{X,Y}(x,y) = P(X = x, Y = y)$$
for all $x, y \in \mathbb{R}$.

Lemma 4.1.2
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X, Y : \Omega \to \mathbb{R}$ be random variables. Then the following are equivalent.

- X and Y are independent.
- $f_{X,Y} = f_X f_Y$.
- There exists functions $g, h : \mathbb{R} \to \mathbb{R}$ such that $f_{X,Y} = gh$.

Proposition 4.1.3 (Law of the Unconscious Staticians) Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \to \mathbb{R}$ be random variables. Let $g : \mathbb{R}^n \to \mathbb{R}$ be a function. Then we have
$$E[g \circ (X_1, \dots, X_n)] = \int_{\mathbb{R}^n} g(x_1, \dots, x_n) f_{(X_1,\dots,X_n)}(x_1, \dots, x_n) \, dx_1 \cdots dx_n$$

4.2 The Expectation and Variance Vector

5 Conditional Probability Theory

5.1 Conditional Probability Density and Mass Functions

Lemma 5.1.1
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X, Y : \Omega \to \mathbb{R}$ be random variables. Suppose that $P(Y = y \mid X = x)$ exists for all $x \in X$ and $y \in Y$. Then $Y \mid X = x : \Omega \to \mathbb{R}$ given by $y \mapsto P(Y = y \mid X = x)$ is a random variable.

Thus it makes sense to talk about its probability density / mass function.

Definition 5.1.2 (Conditional Probability Density and Mass Functions)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X, Y : \Omega \to \mathbb{R}$ be random variables. Suppose that $f_Y(y) > 0$ for some $y \in \Omega$. Define the conditional probability mass function of X given $Y = y$ to be the function

$$f_{X|Y}(x, y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}$$

5.2 Conditional Expectations on an Event

Definition 5.2.1 (Conditional Expectations on an Event)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}$ be a random variable. Let $A \in \mathcal{F}$ be an event such that $P(A) > 0$. Define the conditional expectation of X on A to be

$$E[X \mid A] = \frac{E[X \cdot 1_A]}{P(A)}$$

Lemma 5.2.2
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}$ be a random variable. Let $A \in \mathcal{F}$ be an event such that $P(A) > 0$. Then we have

$$E[X \mid A] = \int_\Omega X \, dP(- \mid A)$$

In other words, conditional expectation is just the definition of expectations applied to conditional probability $(\Omega, \mathcal{F}, P(- \mid A))$.

Lemma 5.2.3
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X, Y : \Omega \to \mathbb{R}$ be random variables. Then we have

$$E[X \mid Y = y] = \int_{x \in \text{im}(X)} x f_{X|Y}(x, y)$$

5.3 Conditional Expectations on Sigma Subalgebras

Definition 5.3.1 (Conditional Expectations on Subalgebras)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}$ be a random variable. Let $\mathcal{H}$ be a σ-subalgebra of $\mathcal{F}$. Define $E[X \mid \mathcal{H}] : \Omega \to \mathbb{R}$ to be a random variable such that the following are true.
- $E[X \mid \mathcal{H}]$ is $\mathcal{H}$-measurable.
- For any $A \in \mathcal{H}$, we have $E[X \cdot 1_A] = E[E[X \mid \mathcal{H}] \cdot 1_A]$

Lemma 5.3.2
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}$ be a random variable. Let $\mathcal{H}$ be a σ-subalgebra of $\mathcal{F}$. Then the random variable $E[X \mid \mathcal{H}]$ exists and is unique up to almost surely equality.

Lemma 5.3.3
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}$ be a random variable. Let $\mathcal{H}$ be a σ-subalgebra of $\mathcal{F}$. Then we have
$$\int_A E[X \mid \mathcal{H}] \, dP = \int_A X \, dP$$
for all $A \in \mathcal{H}$.

Lemma 5.3.4
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $A \in \mathcal{F}$ be an event. Let $\mathcal{A} = \{\emptyset, A, \Omega \setminus A, \Omega\}$. Then we have
$$E[X \mid \mathcal{A}] = E[X \mid A]1_A + E[X \mid \Omega \setminus A]1_{\Omega \setminus A}$$

Thus conditional expectations on σ-subalgebras generalize conditional expectations on events.

Lemma 5.3.5
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X, Y : \Omega \to \mathbb{R}$ be random variables. Let $\mathcal{H}$ be a σ-subalgebra of $\mathcal{F}$. Then the following are true.

- Linearity: If X, Y is $\mathcal{H}$-measurable and $\lambda, \mu \in \mathbb{R}$, then we have $E[\lambda X + \mu Y \mid \mathcal{H}] = \lambda E[X \mid \mathcal{H}] + \mu E[Y \mid \mathcal{H}]$.
- Tower property: If X is $\mathcal{H}$-measurable and $\mathcal{G}$ is a σ-subalgebra of $\mathcal{H}$, then we have $E[E[X \mid \mathcal{H}] \mid \mathcal{G}] = E[X \mid \mathcal{G}]$.
- Stability: If X is $\mathcal{H}$-measurable, then $E[XY \mid \mathcal{H}] = XE[Y \mid \mathcal{H}]$.
- Independence: If $\sigma(X)$ and $\mathcal{H}$ are independent, then $E[X \mid \mathcal{H}] = E[X]$.

Definition 5.3.6 (Conditional Expectation on Random Variables)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X, Y : \Omega \to \mathbb{R}$ be random variables. Define the conditional expectation of X on Y to be
$$E[X \mid Y] = E[X \mid \sigma(Y)]$$

Lemma 5.3.7
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X, Y : \Omega \to \mathbb{R}$ be random variables. Then we have
$$E[Y \mid X](\omega) = E[Y \mid X = X(\omega)]$$

We have the following commutative diagram:
$$\begin{tikzcd}
\Omega \arrow[r, "{E[Y \mid X]}"] \arrow[d, "X"'] & \mathbb{R} \\
\mathbb{R} \arrow[r, "{x \mapsto Y \mid X = x}"'] & \{Z : \Omega \to \mathbb{R} \mid Z \text{ is a random variable }\} \arrow[u, "{E[-]}"']
\end{tikzcd}$$

Theorem 5.3.8 (Law of Total Expectations)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X, Y : \Omega \to \mathbb{R}$ be random variables. Then we have
$$E[X] = E[E[X \mid Y]] = \int_{\text{im}(Y)} E[X \mid Y = y] f_Y(y) \, dy$$

Example 5.3.9
Suppose you roll a fair dice. For each roll, you are paid the face value. If a roll gives $4, 5, 6$, you may roll again. Otherwise, the game stops. What is the expected payoff of the game? (c.f. Green Book of Quant)

Answer
Let $E[X]$ be the expected pay off of a roll. Let Y be the random variable that is the roll of the first

dice. By the law of total expectations, we have

$$\begin{aligned}
E[X] &= E[E[X \mid Y]] \\
&= \sum_{y \in Y} E[X \mid Y = y] f|_Y(y) \\
&= E[X \mid Y = \{1,2,3\}]\frac{1}{2} + E[X \mid Y = \{4,5,6\}]\frac{1}{2} \\
&= 1 + \frac{1}{2}(5 + E[X])
\end{aligned}$$

Solving this gives $E[X] = 7$. ⊛

6 Convergence of Random Variables

6.1 Different Notions of Convergences

Definition 6.1.1 (Convergence in Mean Square) We say that a sequence of random variables $X_1, X_2, \dots$ converges in mean square to a random variable X if

$$\lim_{n\to\infty} E[(X_n - X)^2] = 0$$

Definition 6.1.2 (Convergence in Probability) We say that a sequence of random variables $X_1, X_2, \dots$ converges in probability to a random variable X if for every $\epsilon > 0$, we have

$$\lim_{n\to\infty} P(|X_n - X| > \epsilon) = 0$$

Definition 6.1.3 (Convergence in Distribution) We say that a sequence of random variables $X_1, X_2, \dots$ converges in distribution to a random variable X if

$$\lim_{n\to\infty} F_{X_n}(x) = F_X(x)$$

for every x in the set $C = \{x \in \mathbb{R} : F_X \text{ is continuous at } x\}$.

Proposition 6.1.4 Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_n : \Omega \to \mathbb{R}$ be a sequence of random variables for $n \in \mathbb{N} \setminus \{0\}$. Let $X : \Omega \to \mathbb{R}$ also be a random variable. Then the following are true.

- If X_n converges in mean square to X, then X_n converges in probability to X.
- If X_n converges in probability to X, then X_n converges in distribution to X.

6.2 Law of Large Numbers

Theorem 6.2.1 (Markov Inequality) Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}$ be a random variable. If $E[X] < \infty$, then we have

$$P(|X| \geq a) \leq \frac{E[X]}{a}$$

for any $a > 0$.

Theorem 6.2.2 (Chebyshev Inequality) Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}$ be a random variable. If $E[X^2] < \infty$, then we have

$$P(|X| \geq a) \leq \frac{E[X^2]}{a}$$

for any $a > 0$.

Theorem 6.2.3 (Weak law of large numbers) Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_n : \Omega \to \mathbb{R}$ for $n \in \mathbb{N} \setminus \{0\}$ be a sequence of independently identically distributed random variables with mean μ. Let $S_n = \frac{1}{n}\sum_{i=1}^n X_i$. Then we have

$$\lim_{n\to\infty} P(|S_n - \mu| > \varepsilon) = 0$$

for all $\varepsilon > 0$. In other words, $(S_n)_{n\in\mathbb{N}\setminus\{0\}}$ converges in probability to μ.

Theorem 6.2.4 (Law of large numbers in mean square) Let $X_1, X_2, \dots$ be a sequence of independent random variable, each with mean μ and variance σ^2. Then

$$\lim_{n\to\infty} \frac{X_1 + \cdots + X_n}{n} \to \mu$$

in mean square.

6.3 Central Limit Theorem

Theorem 6.3.1 (Central Limit Theorem) Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_n : \Omega \to \mathbb{R}$ be a sequence of independent and identically distributed random variables for $n \in \mathbb{N} \setminus \{0\}$, each with mean μ and variance $\sigma^2 \neq 0$. Let $S_n = X_1 + \cdots + X_n$. Then the standardized version of S_n,

$$Z_n = \frac{S_n - E(S_n)}{\sqrt{\text{Var}(S_n)}} = \frac{S_n - n\mu}{\sigma\sqrt{n}}$$

converges in distribution as $n \to \infty$ to a Gaussian random variable with mean 0 and variance 1. That is,

$$\lim_{n\to\infty} P(Z_n \leq x) = \lim_{n\to\infty} F_{Z_n}(x) = F_Y(y) = \int_{-\infty}^{x} -\frac{1}{\sqrt{2\pi}} e^{-y^2/2}$$

7 Functional Analysis of Random Variables

7.1

Definition 7.1.1 (The Space of Random Variables)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Define the $\mathbb{R}$-vector space of (real-valued) random variables to be the set

$$L^2(\Omega) = \{X : \Omega \to \mathbb{R} \mid X \text{ is a random variable and } E[X^2] < \infty\}$$

together with pointwise addition and scalar multiplication.

Proposition 7.1.2
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Then $L^2(\Omega)$ is a Hilbert space with innner product given by

$$\langle X, Y\rangle = E[XY]$$